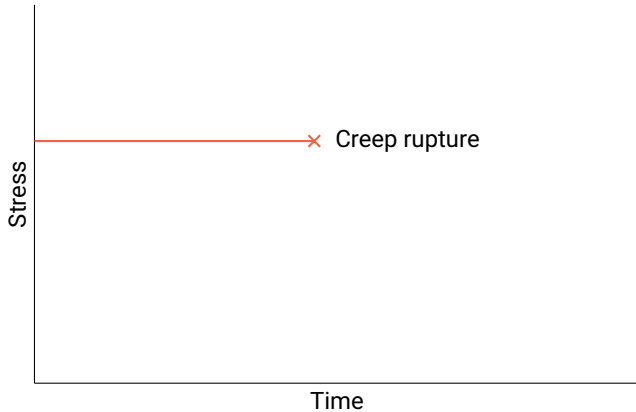


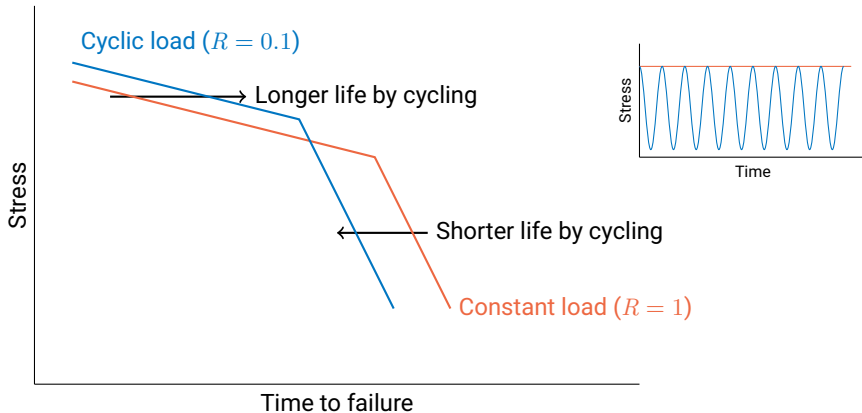
Microscale modeling of creep and fatigue in composites: viscoplasticity, cohesive damage and time homogenization

Dragan Kovačević, Pieter Hofman, Juri Rocha, Frans van der Meer

Cyclic loading has a two-fold effect on the lifetime of polymers

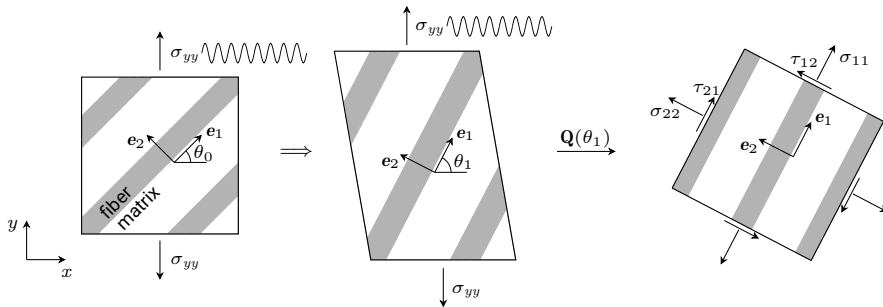


Cyclic loading has a two-fold effect on the lifetime of polymers



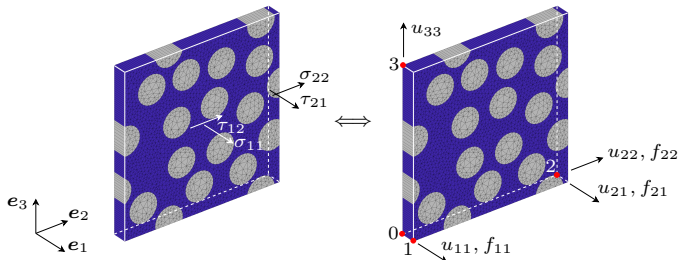
Are these phenomena also observable in composites?

Micromechanical modeling of fatigue in thermoplastics



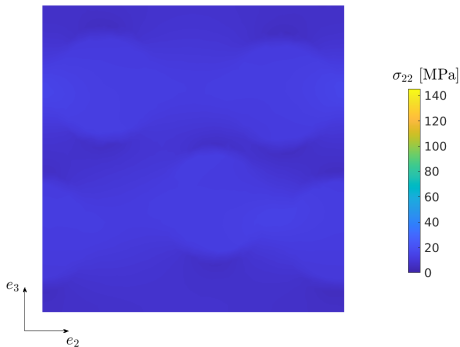
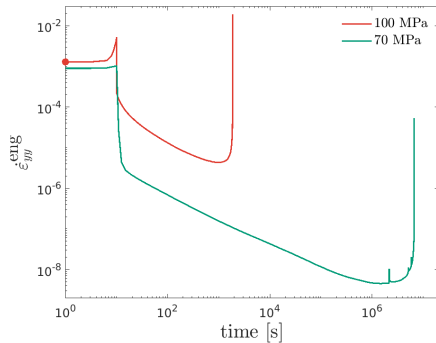
Aim: model cyclic off-axis response

- Thin slice RVE
- BCs for off-axis loading
- Viscous model for the matrix
- Interelement cracks
- Fatigue cohesive zone model



Failure is detected as a quick increase in strain-rate

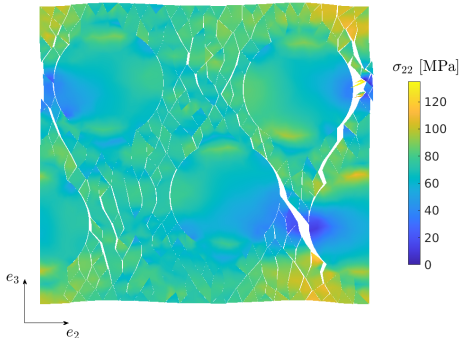
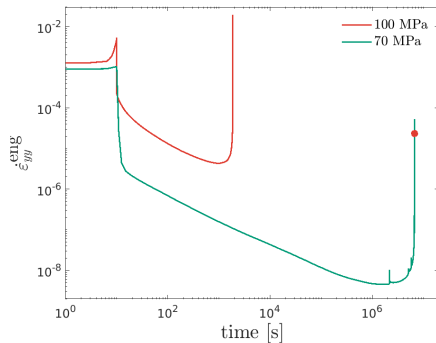
$$\chi = 90^\circ, \sigma_{yy}^{\text{eng}} = 100 \text{ MPa}, R = 0.1, f = 1 \text{ Hz}$$



The simulation terminates when the material cannot carry the applied load

Failure is detected as a quick increase in strain-rate

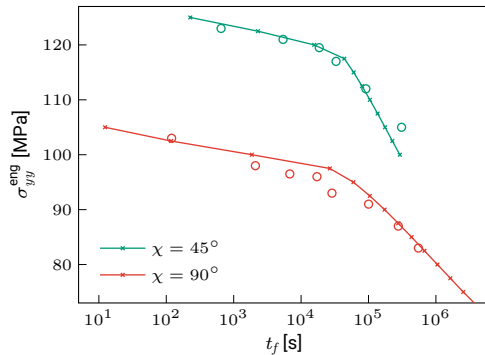
$$\chi = 90^\circ, \sigma_{yy}^{\text{eng}} = 70 \text{ MPa}, R = 0.1, f = 1 \text{ Hz}$$



The simulation terminates when the material cannot carry the applied load

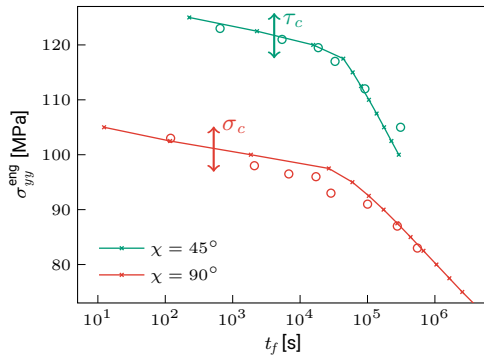
The model is calibrated on a single frequency with two different angles

Calibrating against experiments at 45° and 90° with $R = 0.1$ and $f = 1$ Hz



The model is calibrated on a single frequency with two different angles

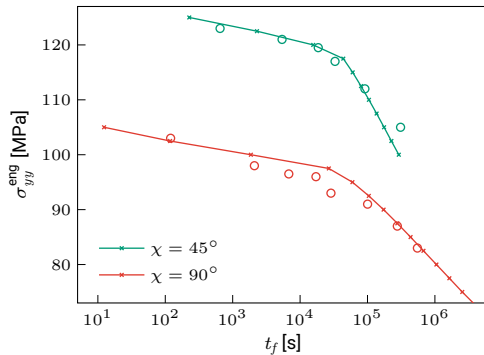
Calibrating against experiments at 45° and 90° with $R = 0.1$ and $f = 1$ Hz



- EGP spectrum from earlier creep simulations
- Cohesive strength to match plasticity regime
 - Tensile strength σ_c for 90°
 - Shear strength τ_c for 45°

The model is calibrated on a single frequency with two different angles

Calibrating against experiments at 45° and 90° with $R = 0.1$ and $f = 1$ Hz



- EGP spectrum from earlier creep simulations
- Cohesive strength to match plasticity regime
 - Tensile strength σ_c for 90°
 - Shear strength τ_c for 45°
- Fatigue parameters for crack growth regime
 - Endurance limit ϵ for offset
 - S-N parameter β for slope

- Transition between regimes can be improved with recalibration of EGP spectrum
- Dependence on fracture mode in the cohesive model needs more attention

Conclusions & discussion

A unified micromechanical framework for modeling creep and fatigue in composites

- ✓ Large deformation nonlinear visco-elastic polymer model: EGP
- ✓ Cohesive zone model with cyclic damage: Dávila
- ✓ Time-homogenization for efficient time stepping with cyclic load

The model reproduces experimentally observed phenomena from off-axis tests

- Two failure regimes with different frequency dependence are distinguished in fatigue simulations
- Creep rupture ($R = 1$) is captured with the same model
- Improvement is possible in dependence on R -value and mode of fracture

[Kovačević et al (2024), *JMPS*]

